INFORMATION SECURITY

Cryptography

Introduction

- Information security means protecting information and information system from unauthorized access, use, modification or destruction(damaged).
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Cryptography
Definition: Use of secret codes to provide integrity/confidentiality of information during transfer and storage

Cryptographic Objectives

- CIA == Confidentiality, Integrity, and Availability
- ***Confidentiality***
  - aims to prevent unauthorized reading of information.
- ***Integrity***
  - detect unauthorized *writing* of information
- ***Availability***
  - Data is available in a *timely manner* when needed

Cryptography

- Cryptography or “secret codes” are a fundamental information security tool.
- **Cryptology** — The art and science of making and breaking “secret codes”
- **Cryptography** — making “secret codes”
- **Cryptanalysis** — breaking “secret codes”

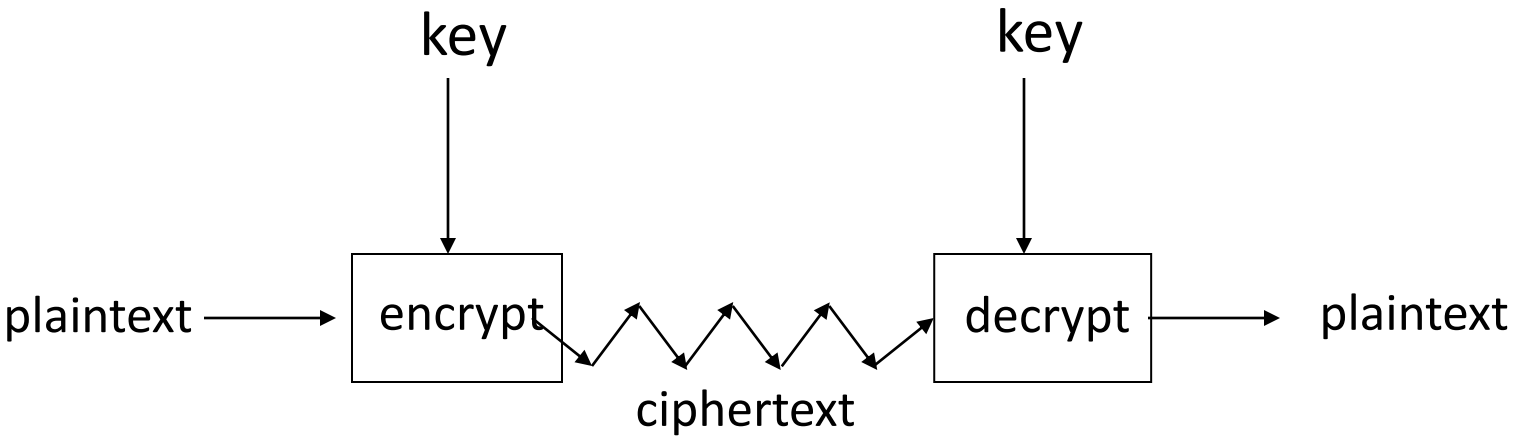
Terminology

- **Plaintext** original message
- **Ciphertext** encrypted or coded message
- **Encryption** convert from plaintext to ciphertext (enciphering)
- **Decryption** restore the plaintext from ciphertext (deciphering)
- **Key** information used in cipher known only to sender/receiver
- **Cipher** a particular algorithm (cryptographic system)

How to Speak Crypto

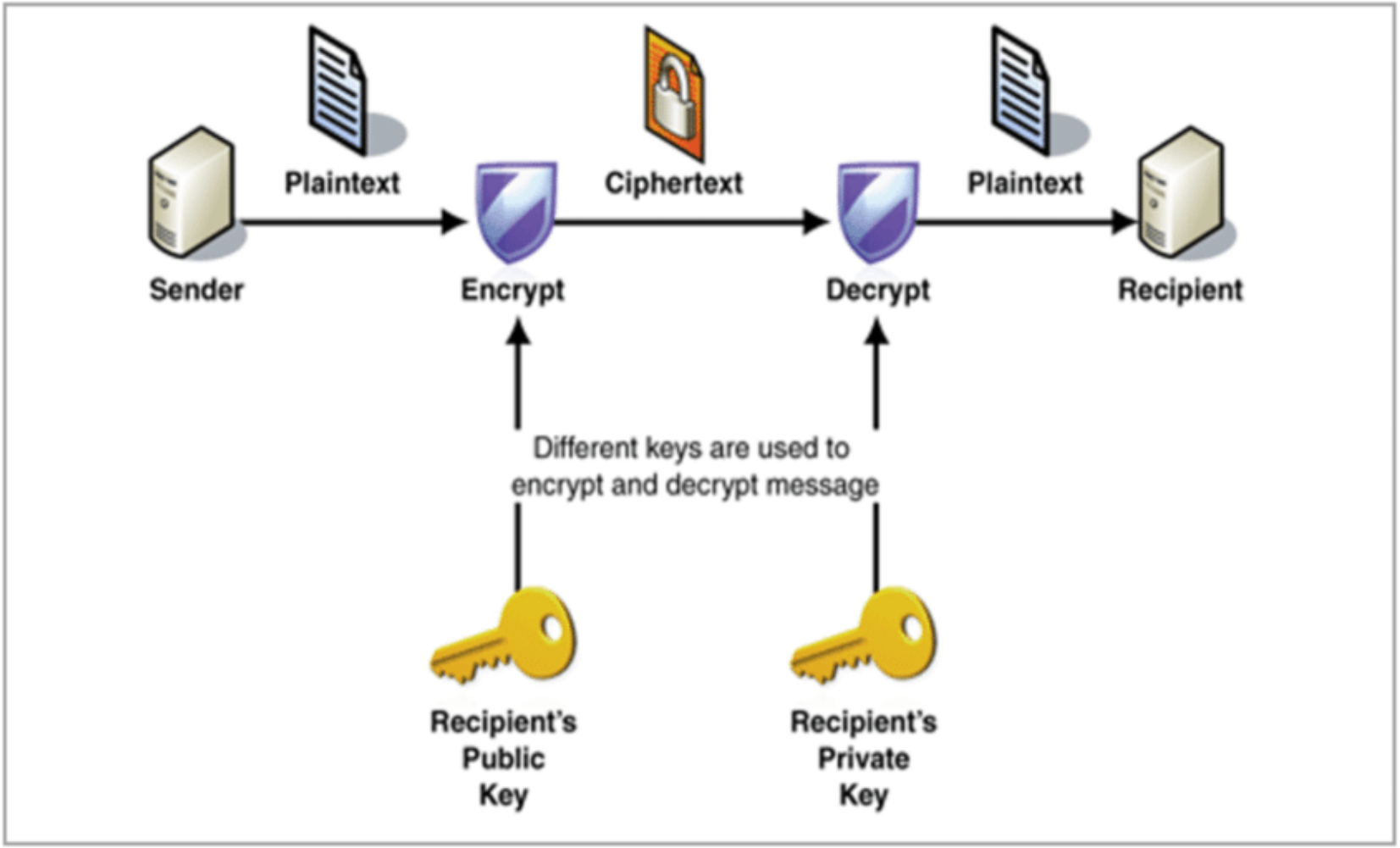
- A *cipher* or *cryptosystem* is used to *encrypt* the *plaintext*
- The result of encryption is *ciphertext*
- We *decrypt* ciphertext to recover plaintext
- A *key* is used to configure a cryptosystem
- A *symmetric key* cryptosystem uses the same key to encrypt as to decrypt
- A *public key (Asymmetric key)* cryptosystem uses a *public key* to encrypt and a *private key* to decrypt

Symmetric key



A generic view of symmetric key crypto

public key


Plaintext
Ciphertext
Plaintext
Sender
Encrypt
Decrypt
Recipient
Different keys are used to
encrypt and decrypt message
Recipient's
Public
Key
Recipient's
Private
Key

Crypto

- Basic assumptions
  - The system is completely known to the attacker
  - Only the key is secret
  - That is, crypto algorithms are not secret
- This is known as **Kerckhoffs' Principle**
- All encryption algorithms are based on two general principles:
  1. **substitution:** in which each element in the plaintext (bit, letter, group of bits or letters) is mapped into another element.
  2. **Transposition:** in which elements in the plaintext are rearranged.
- The fundamental requirement is that no information be lost (that is, that all operations are reversible).

Classic cryptosystems

- We'll examine different classic cryptosystems each of which illustrates some particularly relevant feature:
  - Rail Fence cipher
  - Columnar Transposition cipher
  - Double transposition cipher
  - Simple Substitution cipher
  - Playfair cipher
  - One time pad cipher
  - Vigenere Cipher
  - Affine Cipher
  - Hill Cipher

Rail Fence-Encryption

- To encrypt a message using the Rail Fence Cipher, you have to write your message in zigzag lines across the page, and then read off each row.
- Firstly, you need to have a key, which for this cipher is the number of rows you are going to have.
- You then start writing the letters of the plaintext diagonally down to the right until you reach the number of rows specified by the key.
- You then bounce back up diagonally until you hit the first row again.
- This continues until the end of the plaintext.

Rail Fence cipher

- The Rail Fence cipher works by writing your message on alternate lines across the page, and then reading off each line in turn.
- For example, the plaintext "defend the east wall" is written as shown below, with all spaces removed.

D		F		N		T		E		A		T		A		L
	E		E		D		H		E		S		W		L	

- The ciphertext is then read off by writing the top row first, followed by the bottom row, to get "DFNTEATALEEDHESWL".

Example-Encryption

- For the plaintext we used above, "defend the east wall", with a key of 3, we get the encryption process shown below.



- The ciphertext is read off row by row to get "DNETLEEDHESWLFTAA".

Rail Fence cipher

- The plaintext is written down as a sequence of diagonals and then read off as a sequence of rows. For example, to encipher the message "meet me after the toga party" with a rail fence of depth 2, we write the following:

```
m e m a t r h t g p r y
  e t e f e t e o a a t
```

- The encrypted message is:

MEMATRHTGPRYETEFETEOAAT

Rail Fence-Decryption

- We start by making a grid with as many rows as the key is, and as many columns as the length of the ciphertext.
- We then place the first letter in the top left square, and dashes diagonally downwards where the letters will be.
- When we get back to the top row, we place the next letter in the ciphertext. Continue like this across the row, and start the next row when you reach the end.

Example-Decryption

- For example, if you receive the ciphertext "TEKOOHRACIRMNREATANFTETYTGHH", encrypted with a key of 4:
- you start by placing the "T" in the first square. You then dash the diagonal down spaces until you get back to the top row, and place the "E" here.
- Continuing to fill the top row you get the pattern below.

Example-Decryption

- We have a table with 4 rows because the key is 4, and 28 columns as the ciphertext has length 28.Continuing this row-by-row, we get the successive stages shown below: TEKOOHRACIRMNREATANFTETYTGHH

T						E						K						O						O			
	-				-		-				-		-				-		-				-		-		
		-		-				-		-				-		-				-		-				-	
			-						-						-						-						-

T						E						K						O						O			
	H				R		A				C		I				R		M				N		R		
		-		-				-		-				-		-				-		-				-	
			-						-						-						-						-

T						E						K						O						O			
	H				R		A				C		I				R		M				N		R		
		E		A				T		A				N		F				T		E				T	
			-						-						-						-						-

T						E						K						O						O			
	H				R		A				C		I				R		M				N		R		
		E		A				T		A				N		F				T		E				T	
			Y						T						G						H						H

Plain text: "they are attacking from the north".

Cryptanalysis

- The cipher's key is N, the number of rails. If N is known, the ciphertext can be decrypted by using the previous algorithm.
- Values of N equal to or greater than L(the length of the ciphertext) are not usable, since then the ciphertext is the same as the plaintext.
- Therefore the number of usable keys is low, allowing the brute-force attack of trying all possible keys.
- As a result, the rail-fence cipher is considered weak.

Columnar Transposition-Encryption

- A simple example for a transposition cipher is **columnar transposition cipher**.
- write the message in a rectangle, row by row, and read the message off, column by column.
- permute the order of the columns. The order of the columns then becomes the key to the algorithm.

Key:	4	3	1	2	5	6	7
Plaintext:	a	t	t	a	c	k	p
	o	s	t	p	o	n	e
	d	u	n	t	i	l	t
	w	o	a	m	x	y	z

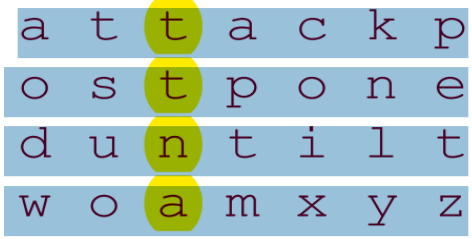
Ciphertext: TTNAAPTMTSUOAODWCOIXKNLYPETZ

Columnar Transposition-decryption

- To decipher it, the recipient has to work out the column lengths by dividing the message length by the key length.
- Then, write the message out in columns again, then re-order the columns by reforming the key word.
- Key: 4312567
- Ciphertext:

TTNA APTM TSUO AODW COIX KNLY PETZ

- Plaintext:



Example 1

- We first pick a keyword for our encryption. We write the plaintext out in a grid where the number of columns is the number of letters in the keyword. We then title each column with the respective letter from the keyword. We take the letters in the keyword in alphabetical order, and read down the columns in this order. If a letter is repeated, we do the one that appears first, then the next and so on.
- **Example**:
- encrypt the message "The tomato is a plant in the nightshade family"
- keyword *tomato*
- Ciphertext: TINESAX / EOAHTFX / HTLTHEY / MAIIAIX / TAPNGDL / OSTNHMX" (where the / tells you where a new column starts).
- The final ciphertext is thus
- "TINES AXEOA HTFXH TLTHE YMAII AIXTA PNGDL OSTNH MX".

T	O	M	A	T	O
5	3	2	1	6	4
T	H	E	T	O	M
A	T	O	I	S	A
P	L	A	N	T	I
N	T	H	E	N	I
G	H	T	S	H	A
D	E	F	A	M	I
L	Y	X	X	X	X

Example 2

- decrypt the ciphertext "ARESA SXOST HEYLO IIAIE XPENG DLLTA HTFAX TENHM WX" given the keyword *potato*.
- There are 42 letters in the ciphertext, and the keyword has six letters, so we need 42 ÷ 6 = 7 row

P	O	T	A	T	O
4	2	5	1	6	3

Example 2

P	O	T	A	T	O
4	2	5	1	6	3
			A		
			R		
			E		
			S		
			A		
			S		
			X		

P	O	T	A	T	O
4	2	5	1	6	3
	O		A		
	S		R		
	T		E		
	H		S		
	E		A		
	Y		S		
	L		X		

P	O	T	A	T	O
4	2	5	1	6	3
	O		A		O
	S		R		I
	T		E		I
	H		S		A
	E		A		I
	Y		S		E
	L		X		X

P	O	T	A	T	O
4	2	5	1	6	3
P	O	T	A	T	O
E	S	A	R	E	I
N	T	H	E	N	I
G	H	T	S	H	A
D	E	F	A	M	I
L	Y	A	S	W	E
L	L	X	X	X	X

- The plaintext :"potatoes are in the nightshade family as well".

Cryptanalysis

Key Length	No. of permutations
2	2
3	6
4	24
5	120
6	720
7	5,040
8	40,320
9	362,880
10	3,628,800
11	39,916,800
12	479,001,600

Cryptanalysis

- The columnar transposition cipher is not the easiest of transposition ciphers to break, but there are statistical properties of language that can be exploited to recover the key.
- To greatly increase the security, a substitution cipher could be employed as well as the transposition.
- A peculiarity of transposition ciphers is that the frequency distribution of the characters will be identical to that of natural text (since no substitutions have been performed, it is just the order that has been mixed up).

Double Transposition Cipher

- To encrypt with a double transposition cipher, we first write the plaintext into an array of a given size and then permute the rows and columns according to specified permutations.
- For example, suppose we write the plaintext **attackatdawn** into a 3 × 4 array:

$$\begin{bmatrix} a & t & t & a \\ c & k & a & t \\ d & a & w & n \end{bmatrix}$$

Double Transposition Cipher

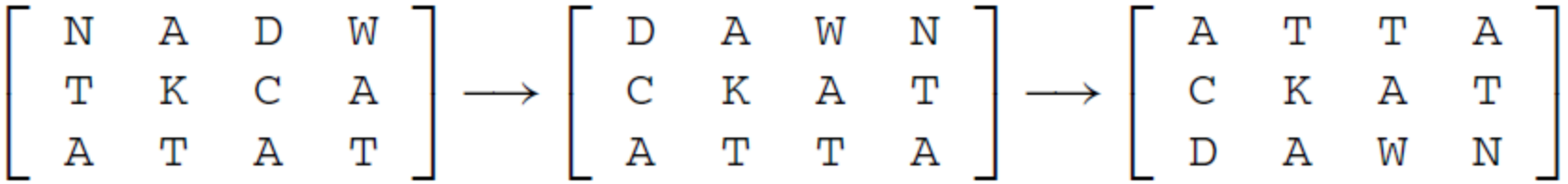
- Now if we transpose (or permute) the rows according to *(*1, 2, 3*)* → *(*3, 2, 1*)* and then transpose the columns according to *(*1, 2, 3, 4*)* → *(*4, 2, 1, 3*)*, we obtain:

$$\begin{bmatrix} a & t & t & a \\ c & k & a & t \\ d & a & w & n \end{bmatrix} \longrightarrow \begin{bmatrix} d & a & w & n \\ c & k & a & t \\ a & t & t & a \end{bmatrix} \longrightarrow \begin{bmatrix} n & a & d & w \\ t & k & c & a \\ a & t & a & t \end{bmatrix}$$

Ciphertext: NADWTKCAATAT

Double Transposition Cipher

- For the double transposition, the key consists of the size of the matrix and the row and column permutations.
- For example, to decrypt *pervious* ciphertext is first put into a 3 × 4 array.
- Then the columns are numbered as *(4, 2, 1, 3)* and rearranged to *(1, 2, 3, 4)*.
- Then the rows are numbered *(3, 2, 1)* and rearranged into *(1, 2, 3)*, as illustrated below



Double Transposition Cipher

Suppose we want to encrypt the plaintext message

attackxatxdawn (where "x" represents a space)

For example, we could put the message above into a 5×3 matrix as below.

	col 1	col 2	col 3
row 1	a	t	t
row 2	a	c	k
row 3	x	a	t
row 4	x	d	a
row 5	w	n	x

Suppose we permute the columns (1,2,3) ⟶ (1,3,2) and the rows (1,2,3,4,5) ⟶ (3,5,1,4,2).

Double Transposition Cipher

	col 1	col 3	col 2
row 3	x	t	a
row 5	w	x	n
row 1	a	t	t
row 4	x	a	d
row 2	a	k	c

Then the ciphertext is

xtawxnattxadakc

PROBLEM

Encrypt the message

we are all together

using a double transposition cipher with 4 rows and 4 columns, using the row permutation

(1, 2, 3, 4) → (2, 4, 1, 3)

and the column permutation

(1, 2, 3, 4) → (3, 1, 2, 4).